



CafeBook — Project Requirements

Cafe Seat Booking & Reservation Management System

Document Date: May 2026 | Version 1.0 | Status: Complete

1. Project Purpose

CafeBook is a web-based table reservation system that allows customers to browse available seats, choose their preferred seating zone, and confirm bookings online — without calling the cafe. Staff and administrators can manage all reservations and tables from a dedicated dashboard.

2. User Roles

Role	Who	What They Can Do
Guest	Any registered customer	Register, log in, search available seats, make a booking, view and cancel their own reservations
Admin	Cafe staff / manager	View all bookings, update booking status, add new tables, mark tables available or unavailable, delete tables

Default admin login is created automatically when the server first starts: **admin@cafe.com / admin123**

3. Customer Features

A Account & Login

- Create account with name, email, phone
- Optional: dietary notes, accessibility need
- Sign in with email and password
- Passwords stored securely (BCrypt)

B Find a Seat

- Choose date, time and number of guests
- Filter by seating zone (6 zones available)
- Only seats that fit the group size are shown
- Already-booked seats are hidden automatically

C Make a Booking

- Select a table from available results
- Add optional special notes or occasion
- Receive a booking reference (e.g. CB-00001)
- Booking starts as **PENDING** status

D My Reservations

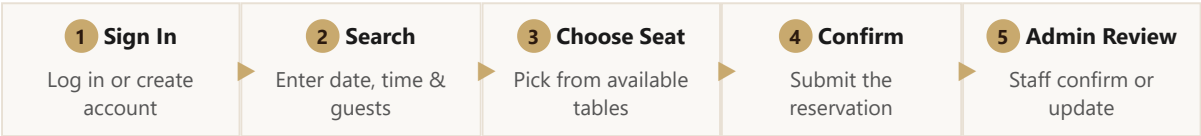
- View all past and upcoming bookings
- See seat number, zone, date, time, status
- Cancel any PENDING or CONFIRMED booking

4. Seating Zones (Pre-configured)

Zone	Description	Tables	Accessible
Window Side	Scenic views with natural light	W1, W2	No
Cosy Corner	Quiet, intimate setting	C1, C2	No
Outdoor Patio	Fresh air open seating	P1, P2	Yes

Private Nook	Semi-private booth seating	N1, N2	No
Main Floor	Central, lively atmosphere	M1, M2	Yes (M1)
Lounge	Relaxed low-seating area	L1, L2	No

5. How Booking Works (Step by Step)



6. Booking Status Values

PENDING — Just created, awaiting admin confirmation

CONFIRMED — Approved by admin, seat is reserved

COMPLETED — Customer visited and finished

CANCELLED — Cancelled by customer or admin

NO_SHOW — Customer did not arrive

7. Admin Dashboard Features

1 Booking Management

- View all bookings from all customers
- Filter by status, date, or name/email
- Change booking status with a dropdown
- See stats: Total, Confirmed, Pending, Today

2 Table Management

- Add new tables with seat range and type
- Table types: Standard, Group, Private, Merged
- Mark any table available or unavailable
- Delete tables from the system

8. Technical Requirements

> Backend Stack

Java 21

Spring Boot 3.2

Spring Security

Spring Data JPA

H2 Database

JWT Auth

Maven

> Frontend Stack

HTML 5

CSS 3

Vanilla JavaScript

Google Fonts

Fetch API

Item	Detail
Server port	http://localhost:8080
Database	H2 file-based (auto-created at database/cafebook.mv.db)
Authentication	JWT token stored in browser localStorage, expires in 24 hours
Password security	BCrypt hashing — plain-text passwords are never stored
API style	REST — JSON request and response
Startup	Double-click Start CafeBook.bat — browser opens automatically after 18 seconds
No install needed	H2 database is embedded — no separate database server required

9. Non-Functional Requirements

Security

Usability

- Admin pages blocked from regular users
- All protected API calls require a valid JWT token
- Passwords never sent or stored as plain text

- Responsive design works on mobile and desktop
- Toast notifications confirm every action
- Form errors shown clearly near the form

Reliability

- Seed data (admin + zones + tables) is created once on first run
- Duplicate bookings for the same seat/time are prevented
- Database persists across restarts (file-based H2)

Maintainability

- 28 source files total — small and readable
- Single bat file to start the whole application
- Clean folder structure: frontend / backend / database